Lab No. 8

Design of Sequential circuits

1. Analyze the given state table of a synchronous sequential circuit and perform state reduction. Using the reduced state table, design and ~~stimulate~~ simulate the sequential circuit using D flip-flops.

Present State	Next State		Output	
	$x=0$	$x=1$	$x=0$	$x=1$
a	f	b	0	0
b	d	c	0	0
c	f	e	0	0
d	g	a	1	0
e	d	c	0	0
f	f	b	1	1
g	g	h	0	1
h	g	a	1	0

a	f	b c	0	0
b	d	c	0	0
~~c~~	~~f~~	~~e~~ b	~~0~~	~~0~~
d	g	~~a~~ h	1	0
~~e~~	~~d~~	~~c~~	~~0~~	~~0~~
f	f	b	1	1
g	g	h	0	1
~~h~~	~~g~~	~~a~~	~~1~~	~~0~~

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Reduced State Table

Present state	Next state		Output	
	$x=0$	$x=1$	$x=0$	$x=1$
a	f	c	0	0
b	d	c	0	0
d	g	h	1	0
f	f	b	1	1
g	g	h	0	1

Assigning 3 bit binary values to states

Reduced Table

Present Table	Next State		Output	
~~a = 000~~	$x=0$	$x=1$	$x=0$	$x=1$
a = 000	011	001	0	0
b = 001	010	000	0	0
d = 010	100	000	1	0
f = 011	011	001	1	1
g = 100	100	010	0	1

~~Excita~~

Excitation Table

Present state			Input	Next state						Output
A	B	C	X	A	B	C	D_A	D_B	D_C	Y
0	0	0	0	0	0	1	0	1	1	0
0	0	0	1	0	0	1	0	0	1	0
0	0	1	0	0	1	0	0	1	0	0
0	0	1	1	0	0	0	0	0	0	0
0	1	0	0	1	0	0	1	0	0	1
0	1	0	1	0	0	0	0	0	0	0
0	1	1	0	0	1	1	0	1	1	1
0	1	1	1	0	0	1	0	0	1	1
1	0	0	0	1	0	0	1	0	0	0
1	0	0	1	0	1	0	0	1	0	1

AX \ CX	00	01	11	10
00	0			
01	1			
11	X	X	X	X
10	1		X	X

$DA = BC'X' + AX'$

AB \ CX	00	01	11	10
00	1			1
01				1
11	X	X	X	X
10		1	X	X

$DB = A'B'X' + AX + CX'$

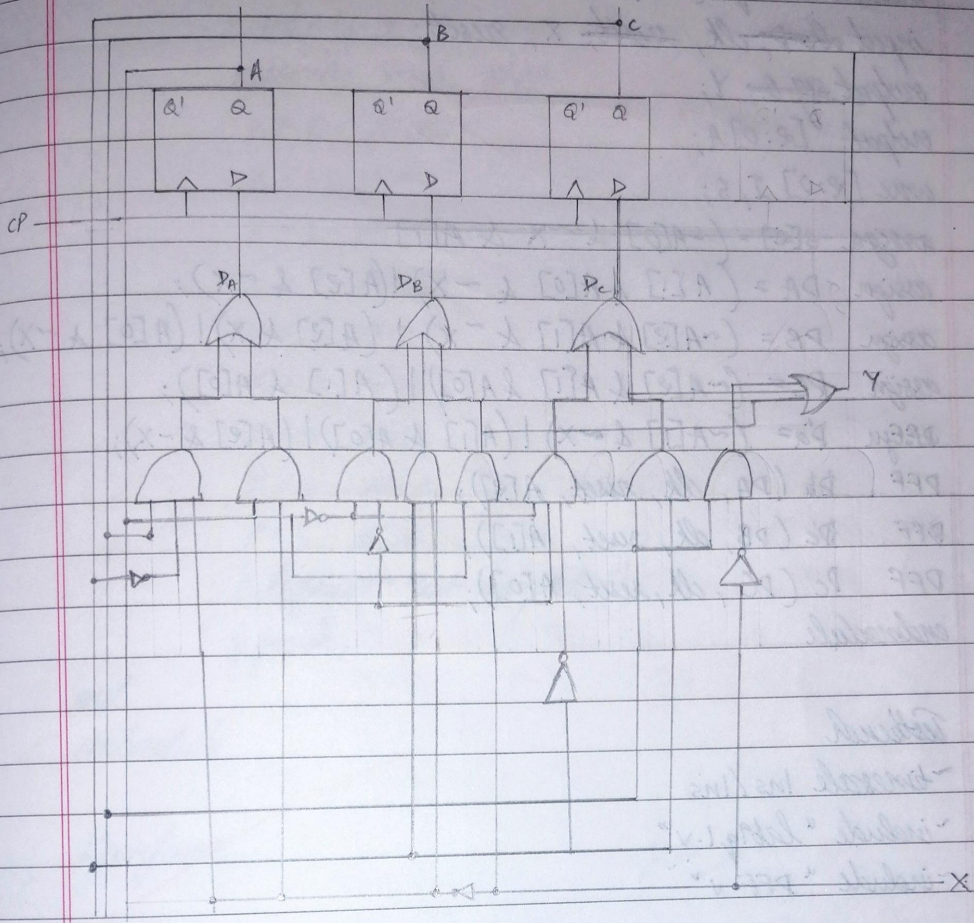
AB \ CX	00	01	11	10
00				
01	1		1	1
11	X	X	X	X
10		1	X	X

$Y = BX' + BC + AX$

AB \ CX	00	01	11	10
00	1	1		
01			1	1
11	X	X	X	X
10			X	X

$DC = A'B'C' + BC$

Circuit



Code

```
module dff(D, Q, clk, reset);
input D, clk, reset;
output reg Q;
always @ (posedge clk or posedge reset)
begin
if (reset)
Q <= 1'b0;
else
Q <= D;
end
endmodule
```

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```
module lab9q1 (clk, X, Y, reset, A);
input ~~clk D,~~ clk, ~~reset,~~ X, reset;
output ~~reg Q;~~ Y;
output [2:0] A;
wire [2:0] I, S;
~~assign S[2] = (~A[0] & ~X & A[1]~~
assign DA = (A[1] & A[0] & ~X) | (A[2] & ~X);
assign DB = (~A[2] & A[1] & ~X) | (A[2] & X) | (A[0] & ~X);
assign DC = (~A[2] & A[1] & A[0]) | (A[1] & A[0]);
assign Y = (~A[1] & ~X) | (A[1] & A[0]) | (A[2] & ~X);
DFF Da (DA, clk, reset, A[2]);
DFF Db (DB, clk, reset, A[1]);
DFF Dc (DC, clk, reset, A[0]);
endmodule
```

Testbench

```
`timescale 1ns/1ns
`include "lab9q1.v"
`include "DFF.v"

module lab9q1_tb;
reg X, clk;
reg reset;
wire Y;
wire [2:0] A;

lab9q1 dup (X, clk, reset, Y, A);
initial begin
    $dumpfile("lab9q1.vcd");
    $dumpvars(0, lab9q1_tb);
    clk = 0;
    forever #10; clk = ~clk;
end
```

```
module lab9q1 (clk, X, Y, reset, A);
input clk, X, reset;
output Y;
output [2:0] A;
wire [2:0] I, S;
assign DA = (A[1] & A[0] & ~X) | (A[2] & ~X);
assign DB = (~A[2] & A[1] & ~X) | (A[2] & X) | (A[0] & ~X);
assign DC = (~A[2] & A[1] & A[0]) | (A[1] & A[0]);
assign Y = (~A[1] & ~X) | (A[1] & A[0]) | (A[2] & ~X);
DFF   Da (DA, clk, reset, A[2]);
DFF   Db (DB, clk, reset, A[1]);
DFF   Dc (DC, clk, reset, A[0]);
endmodule
```

Testbench

```
`timescale 1ns/1ns
`include "lab9q1.v"
`include "DFF.v"

module lab9q1_tb;
reg X, clk;
reg reset;
wire Y;
wire [2:0] A;

lab9q1 dup (X, clk, reset, Y, A);
initial begin
    $dumpfile("lab9q1.vcd");
    $dumpvars(0, lab9q1_tb);
    clk=0;
    forever #10; clk=~clk;
end
```

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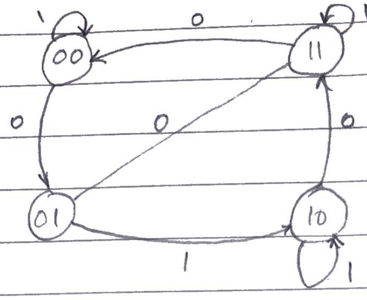
```
initial begin
        reset=1; x=0; #20;
        reset=0; x=0; #20;
          x=0; #20;
          x=1; #20;
          x=1; #20;
          x=1; #20;
          x=0; #20;
          x=0; #20;
          x=0; #20;
          x=1; #20;
          x=1; #20;
          $display("Test complete");
          $finish;
end
endmodule
```

Waveform

2. Design the synchronous sequential circuit for the following state diagram using TFF. Write the verilog code for the derived circuit.



State Table

Present Table state		Next State			
		X=0		X=1	
A	B	A	B	A	B
0	0	0	1	0	0
0	1	1	1	1	0
1	0	1	1	1	0
1	1	0	0	1	1

Excitation Table

i/p of combination circuits			Next State		o/p of circuits flipflop i/p's	
A	B	X	A	B	T_A	T_B
0	0	0	0	1	0	1
0	0	1	0	0	0	0
0	1	0	1	1	1	0
0	1	1	1	0	1	1
1	0	0	1	1	0	1
1	0	1	1	0	0	0
1	1	0	0	0	1	1
1	1	1	1	1	0	0

A \ BX	00	01	11	10
0			1	1
1				1

T_A

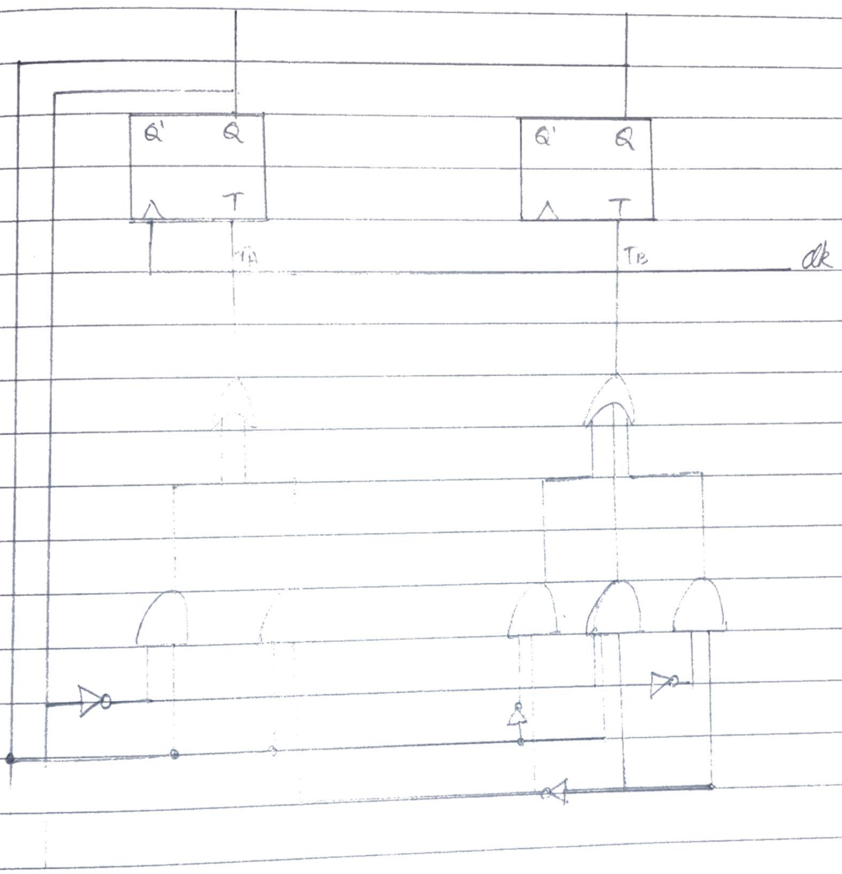
$T_A = A'B + BX'$

A \ BX	00	01	11	10
0	1		1	
1	1			1

T_B

$T_B = B'X' + A'B + AX'$

Circuit



Code

```
module lab9q2(M, clk, reset, A);
input M, clk, reset
output [1:0] A;
assign Ta = (~A[1] & A[0]) | (A[0] & ~x);
assign Tb =   (~A[0] & ~x) | (~A[1] & A[0] & ~x) | (A[1] & ~x);
TFF  TA (Ta, clk, reset, A[1]);
TFF  TA (Tb, clk, reset, A[0]);
endmodule
```

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Testbench

```
~timescale 1ns/1ns
~include "lab9q1.v"

module lab9q2_tb;
reg x, clk;
reg reset;
wire [1:0] A;

~~reg~~ lab9q2 dup(x, clk, reset, A);
initial begin
        $dumpfile ("lab9q2_tb.vcd");
        $dumpvars(0, lab9q2_tb);
        clk=0;
        forever #10 clk =~clk;
end
initial begin
      reset=1;  x=0;  #20;
      reset=0;  x=0;  #20;
      x=0;  #20;
      x=1;  #20;
      x=1;  #20;
      x=0;  #20;
      x=1;  #20;
```

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```
X=0; #20;
X=0; #20;
X=0; #20;
X=0; #20;
$display("Test complete");
$finish;
endmodule
```

Waveform